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Set Theory and Metric Spaces (MAT304)

Unit II: Concepts in Metric Spaces

Complete Lecture Notes

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1 Introduction to Metric Spaces

1.1 The Central Idea: Abstracting Distance

In Euclidean geometry, we have an intuitive notion of distance. The metric space concept—developed by Maurice Fréchet in 1906—abstracts this notion and asks: *What are the minimal properties a "distance" function must satisfy to be useful?* The answer requires only four axioms.

1.2 Definition of a Metric Space

Definition 1.1 (Metric Space)**.** Let X be a non-empty set. A function $d : X \times X \to [0, \infty)$ is called a **metric** (or **distance function**) if for all $x, y, z \in X$:

(i) **Non-negativity and Identity of Indiscernibles**: $d(x, y) = 0 \iff x = y$.

(ii) **Symmetry**: $d(x, y) = d(y, x)$.

(iii) **Triangle Inequality**: $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is called a **metric space**.

Remark 1.1 (The Triangle Inequality Intuition)**.** The triangle inequality captures the fundamental geometric truth that *the shortest path between two points is the direct one.* Any detour through a third point y cannot be shorter than going directly from x to z.

Theorem 1.1 (Variant of the Triangle Inequality)**.** For any x, y, z in a metric space (X, d):

$$|d(x, z) - d(y, z)| \leq d(x, y).$$

Proof. By the triangle inequality, $d(x, z) \leq d(x, y) + d(y, z)$, so $d(x, z) - d(y, z) \leq d(x, y)$. Interchanging x and y, we get $d(y, z) - d(x, z) \leq d(x, y)$. Combining both:

$$-(d(x, y)) \leq d(x, z) - d(y, z) \leq d(x, y),$$

which is exactly $|d(x, z) - d(y, z)| \leq d(x, y)$. □

Remark 1.2. This variant says: *the distance function is 1-Lipschitz in each variable.* It is used frequently to prove the continuity of the distance function $d : X \times X \to \mathbb{R}$.

1.3 Rich Examples of Metric Spaces

Example 1.1 (The Real Line)**.** $X = \mathbb{R}$, $d(x, y) = |x - y|$.

Example 1.2 (Euclidean n-Space)**.** $X = \mathbb{R}^n$, $d_2(x, y) = \left(\sum_{k=1}^{n} |x_k - y_k|^2\right)^{1/2}$.

Example 1.3 (The Taxicab Metric)**.** $X = \mathbb{R}^n$, $d_1(x, y) = \sum_{k=1}^{n} |x_k - y_k|$. This is the **taxicab** or **Manhattan** metric.

Example 1.4 (The Supremum Metric)**.** $X = \mathbb{R}^n$, $d_\infty(x, y) = \max_{1 \leq k \leq n} |x_k - y_k|$. This is the **chessboard** metric.

Example 1.5 (The Discrete Metric)**.** For any set X:

$$d(x,y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

Every distinct pair of points is at distance exactly 1.

Example 1.6 (Matrix Spaces)**.** On $M_{m\times n}(\mathbb{R})$ with entries $A = (a_{ij})$, $B = (b_{ij})$:

- **Frobenius metric**: $d_F(A,B) = \left(\sum_{i,j} |a_{ij} - b_{ij}|^2\right)^{1/2}$.
- **Operator norm metric**: $d_{\text{op}}(A,B) = \sup_{\|x\|\leq 1} \|(A-B)x\|_2$.
- **Entrywise max metric**: $d_{\max}(A,B) = \max_{i,j} |a_{ij} - b_{ij}|$.

For $M_{2\times 2}(\mathbb{R})$, the Frobenius metric is just the Euclidean metric on $\mathbb{R}^4$:

$$d_F\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}\right) = \sqrt{(a_1-a_2)^2 + (b_1-b_2)^2 + (c_1-c_2)^2 + (d_1-d_2)^2}.$$

Example 1.7 (Function Spaces)**.** On $C([a,b])$, the set of continuous real-valued functions:

- **Supremum metric**: $d_\infty(f,g) = \sup_{a\leq x\leq b} |f(x) - g(x)|$.
- L^1 **metric**: $d_1(f,g) = \int_a^b |f(x) - g(x)|\, dx$.
- L^2 **metric**: $d_2(f,g) = \left(\int_a^b |f(x) - g(x)|^2\, dx\right)^{1/2}$.

1.4 Sequence Spaces ℓ^p

Definition 1.2 (The ℓ^p Spaces)**.** For $1 \leq p < \infty$, define

$$\ell^p = \left\{ (x_n)_{n=1}^\infty : x_n \in \mathbb{R},\ \sum_{n=1}^\infty |x_n|^p < \infty \right\},$$

with the metric

$$d_p(x,y) = \left(\sum_{n=1}^\infty |x_n - y_n|^p\right)^{1/p}.$$

For $p = \infty$, define ℓ^∞ as the space of all bounded real sequences, with

$$d_\infty(x,y) = \sup_{n\in\mathbb{N}} |x_n - y_n|.$$

Remark 1.3. The triangle inequality for d_p is the **Minkowski inequality**, which states that for $x, y \in \ell^p$:

$$\left(\sum_{n=1}^\infty |x_n + y_n|^p\right)^{1/p} \leq \left(\sum_{n=1}^\infty |x_n|^p\right)^{1/p} + \left(\sum_{n=1}^\infty |y_n|^p\right)^{1/p}.$$

This is the infinite-dimensional analogue of the Minkowski inequality in $\mathbb{R}^n$.

Theorem 1.2 (Containment Relations of ℓ^p Spaces)**.** If $1 \leq p < q \leq \infty$, then

$$\ell^p \subsetneq \ell^q.$$

Thus we have the chain:

$$\ell^1 \subsetneq \ell^2 \subsetneq \ell^3 \subsetneq \cdots \subsetneq \ell^\infty.$$

Proof. Let $x = (x_n) \in \ell^p$. Since $\sum_{n=1}^{\infty} |x_n|^p < \infty$, the terms must satisfy $x_n \to 0$. Hence there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n| \leq 1$.

Since $q > p$ and $|x_n| \leq 1$ for $n \geq N$, we have $|x_n|^q \leq |x_n|^p$. Therefore,

$$\sum_{n=1}^{\infty} |x_n|^q = \sum_{n=1}^{N-1} |x_n|^q + \sum_{n=N}^{\infty} |x_n|^q \leq \sum_{n=1}^{N-1} |x_n|^q + \sum_{n=N}^{\infty} |x_n|^p < \infty.$$

So $x \in \ell^q$. This proves $\ell^p \subset \ell^q$.

Strictness: Consider the sequence $x_n = \frac{1}{n^{1/p}}$. Then

$$\sum_{n=1}^{\infty} |x_n|^p = \sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

so $x \notin \ell^p$. However, for $q > p$:

$$\sum_{n=1}^{\infty} |x_n|^q = \sum_{n=1}^{\infty} \frac{1}{n^{q/p}},$$

which converges because $q/p > 1$ (a p-series with exponent > 1). So $x \in \ell^q \setminus \ell^p$, proving strict containment. □

Remark 1.4 (Intuition)**.** The containment $\ell^p \subsetneq \ell^q$ for $p < q$ tells us that **higher p means a weaker (more permissive) condition**. Being in ℓ^1 requires absolute summability (the strongest condition), while being in ℓ^∞ only requires boundedness (the weakest condition).

Example 1.8. The harmonic sequence $\left(\frac{1}{n}\right)_{n\geq 1}$ is in ℓ^2 (since $\sum 1/n^2 < \infty$) but not in ℓ^1 (since $\sum 1/n = \infty$). This demonstrates $\ell^1 \subsetneq \ell^2$.

Example 1.9. The sequence $(1)_{n\geq 1}$ (all ones) is in ℓ^∞ (bounded by 1) but not in any ℓ^p for finite p (since $\sum 1 = \infty$). This demonstrates $\ell^p \subsetneq \ell^\infty$ for all finite p.

1.5 The Unit Ball in $\mathbb{R}^2$ with Different Metrics

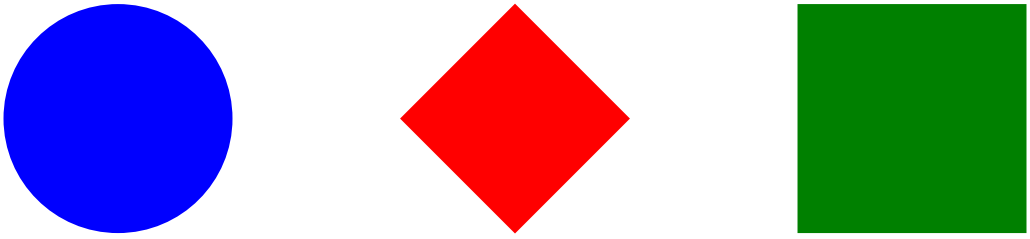
The unit ball at the origin is $B_1(0) = \{x \in \mathbb{R}^2 : d(x, 0) < 1\}$.

(a) **Euclidean (d_2)**: $x^2 + y^2 < 1$ — an open **disk**.

(b) **Taxicab (d_1)**: $|x| + |y| < 1$ — a **diamond** with vertices at $(\pm 1, 0), (0, \pm 1)$.

(c) **Supremum (d_∞)**: $\max\{|x|, |y|\} < 1$ — an axis-aligned **square** $(-1, 1) \times (-1, 1)$.

(d) **Discrete**: $B_r(0) = \{0\}$ if $r \leq 1$, and X if $r > 1$.



Euclidean (d_2) Taxicab (d_1) Supremum (d_∞)

Remark 1.5 (The Shape Matters)**.** The same point $(0.8, 0.6)$ has:

- d_2-norm $= \sqrt{0.64 + 0.36} = 1$ (on the boundary of the Euclidean ball).
- d_1-norm $= 0.8 + 0.6 = 1.4 > 1$ (outside the taxicab ball).
- d_∞-norm $= \max(0.8, 0.6) = 0.8 < 1$ (inside the supremum ball).

So the same point can be on the boundary of one unit ball, outside another, and inside a third. This is why the choice of metric matters.

2 Open and Closed Balls; Neighbourhoods

2.1 Definitions

Definition 2.1 (Open Ball)**.** $B_r(x) = \{y \in X : d(x, y) < r\}$.

Definition 2.2 (Closed Ball)**.** $\overline{B}_r(x) = \{y \in X : d(x, y) \leq r\}$.

Definition 2.3 (Sphere)**.** The **sphere** of radius r centered at x is $S_r(x) = \{y \in X : d(x, y) = r\}$.

2.2 Examples of Open and Closed Balls

Example 2.1 (Balls in $\mathbb{R}$)**.** $B_r(x) = (x - r, x + r)$; $\overline{B}_r(x) = [x - r, x + r]$; $S_r(x) = \{x - r, x + r\}$.

Example 2.2 (Balls in $\mathbb{R}^2$)**.**

- **Euclidean**: $B_r(x_0, y_0)$ is an open disk; $\overline{B}_r(x_0, y_0)$ is a closed disk.
- **Taxicab**: open/closed diamonds.
- **Supremum**: open/closed axis-aligned squares.

Example 2.3 (Balls in the Discrete Metric)**.** $B_r(x) = \{x\}$ if $0 < r \leq 1$, and $B_r(x) = X$ if $r > 1$. Similarly $\overline{B}_r(x) = \{x\}$ if $r < 1$, and $\overline{B}_r(x) = X$ if $r \geq 1$.

Example 2.4 (Balls in Matrix Space with Frobenius Metric)**.** On $M_{2\times 2}(\mathbb{R})$:

$$B_r(0) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a^2 + b^2 + c^2 + d^2 < r^2 \right\}.$$

This is a 4-dimensional open ball. The sphere $S_r(0)$ consists of matrices with Frobenius norm exactly r.

Example 2.5 (Balls in Function Space with Supremum Metric)**.** In $C([0,1])$ with d_∞:

$$B_r(f) = \{g \in C([0,1]) : \sup_{x\in[0,1]} |g(x) - f(x)| < r\}.$$

This consists of all continuous functions whose graphs lie strictly within a **tube** of radius r around the graph of f.

Example 2.6 (Balls in ℓ^p)**.** In ℓ^1: $B_r(0) = \{x : \sum_n |x_n| < r\}$. In ℓ^2: $B_r(0) = \{x : \sum_n |x_n|^2 < r^2\}$. In ℓ^∞: $B_r(0) = \{x : \sup_n |x_n| < r\}$.

2.3 Neighbourhoods

Definition 2.4 (Neighbourhood)**.** A set $N \subset X$ is a **neighbourhood** of $x \in X$ if there exists $r > 0$ with $B_r(x) \subset N$.

Remark 2.1. A neighbourhood need not be open. Any set containing an open ball around x is a neighbourhood.

3 Open Sets

3.1 Definition and Basic Examples

Definition 3.1 (Open Set)**.** $U \subset X$ is **open** if for every $x \in U$, there exists $r > 0$ with $B_r(x) \subset U$.

Example 3.1 (Open Sets in $\mathbb{R}$)**.** Open intervals (a,b), $(-\infty,b)$, (a,∞), $\mathbb{R}$, and $\emptyset$. $\mathbb{Q}$ is not open. Singletons are not open.

Example 3.2 (Discrete Metric)**.** Every subset is open (since $B_1(x) = \{x\} \subset A$ for any $A \ni x$).

3.2 Open Balls are Open Sets

Theorem 3.1. Every open ball $B_r(x)$ is an open set.

Proof. Let $y \in B_r(x)$. Let $t = d(x,y) < r$ and $s = r - t > 0$. For $z \in B_s(y)$:

$$d(x,z) \le d(x,y) + d(y,z) < t + s = r,$$

so $z \in B_r(x)$. Hence $B_s(y) \subset B_r(x)$. Thus every point of $B_r(x)$ has a ball contained in $B_r(x)$, so $B_r(x)$ is open. □

3.3 Properties of Open Sets

Theorem 3.2 (Arbitrary Unions)**.** The union of any collection of open sets is open.

Proof. Let $U = \bigcup_{\alpha\in I} U_\alpha$ with U_α open. For $x \in U$, $x \in U_\alpha$ for some α. Since U_α is open, $\exists r > 0$ with $B_r(x) \subset U_\alpha \subset U$. So U is open. □

Theorem 3.3 (Finite Intersections)**.** The intersection of finitely many open sets is open.

Proof. Let $U = \bigcap_{k=1}^{n} U_k$. For $x \in U$, for each k there is $r_k > 0$ with $B_{r_k}(x) \subset U_k$. Let $r = \min\{r_1, \ldots, r_n\} > 0$. Then $B_r(x) \subset U_k$ for all k, so $B_r(x) \subset U$. Hence U is open. □

Remark 3.1 (Infinite Intersections Fail)**.** In $\mathbb{R}$, $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$, which is not open.

Theorem 3.4 (Union of Open Balls)**.** A set $U \subset X$ is open iff U is a union of open balls.

Proof. ($\Rightarrow$) If U is open, for each $x \in U$ choose $r_x > 0$ with $B_{r_x}(x) \subset U$. Then $U = \bigcup_{x \in U} B_{r_x}(x)$.

($\Leftarrow$) A union of open balls is open by Theorem 3.2 and Theorem 3.3. □

Theorem 3.5 (Characterization via Neighbourhoods)**.** U is open iff for every $x \in U$, U is a neighbourhood of x.

Proof. Immediate from the definitions. □

3.4 The Interior of a Set

Definition 3.2 (Interior Point and Interior)**.** x is an **interior point** of A if $\exists r > 0$ with $B_r(x) \subset A$. The **interior** $\operatorname{int}(A)$ (or A°) is the set of all interior points of A.

Theorem 3.6 (Properties of Interior)**.** (i) $\operatorname{int}(A)$ is open.

(ii) $\operatorname{int}(A) \subset A$.

(iii) $\operatorname{int}(A)$ is the largest open set contained in A.

(iv) A is open iff $A = \operatorname{int}(A)$.

Proof. **(i)** Let $x \in \operatorname{int}(A)$. There is $r > 0$ with $B_r(x) \subset A$. Since $B_r(x)$ is open, for every $y \in B_r(x)$, y is also an interior point of A. Thus $B_r(x) \subset \operatorname{int}(A)$. Hence $\operatorname{int}(A)$ is open.

(ii) If $x \in \operatorname{int}(A)$, then $B_r(x) \subset A$ for some r, and in particular $x \in A$ (since $d(x, x) = 0 < r$). So $\operatorname{int}(A) \subset A$.

(iii) Let $V \subset A$ be open. Take $x \in V$. Since V is open, there exists $r > 0$ with $B_r(x) \subset V \subset A$. Thus $x \in \operatorname{int}(A)$. So $V \subset \operatorname{int}(A)$. Combined with (i), $\operatorname{int}(A)$ is the largest open set inside A.

(iv) If A is open, every $x \in A$ is an interior point, so $A \subset \operatorname{int}(A)$. Combined with (ii), $A = \operatorname{int}(A)$. Conversely, if $A = \operatorname{int}(A)$, then A is open by (i). □

Example 3.3 (Interiors in $\mathbb{R}$)**.** $\operatorname{int}((0,1)) = (0,1)$; $\operatorname{int}([0,1]) = (0,1)$; $\operatorname{int}(\mathbb{Q}) = \emptyset$; $\operatorname{int}(\mathbb{Z}) = \emptyset$.

4 Closed Sets and Limit Points

4.1 Definition of Closed Sets

Definition 4.1 (Closed Set)**.** $F \subset X$ is **closed** if $X \setminus F$ is open.

Remark 4.1. $\emptyset$ and X are clopen (both closed and open).

4.2 Limit Points

Definition 4.2 (Limit Point)**.** x is a **limit point** of A if every neighbourhood of x contains a point of $A \setminus \{x\}$. Equivalently, for every $r > 0$, $B_r(x) \cap (A \setminus \{x\}) \neq \emptyset$.

Example 4.1 (Limit Points in $\mathbb{R}$)**.** Limit points of $(0, 1)$ are $[0, 1]$; of $\mathbb{Q}$ are $\mathbb{R}$; of $\mathbb{Z}$ are $\emptyset$; of $\{1/n\}$ are $\{0\}$.

Definition 4.3 (Isolated Point)**.** $x \in A$ is **isolated** if it is not a limit point of A. Equivalently, $\exists r > 0$ with $B_r(x) \cap A = \{x\}$.

4.3 Sequential Characterization of Limit Points

Theorem 4.1 (Sequences Detect Limit Points)**.** x is a limit point of A iff there exists a sequence $(x_n) \subset A \setminus \{x\}$ such that $x_n \to x$.

Proof. ($\Rightarrow$) For each n, $B_{1/n}(x) \cap (A \setminus \{x\}) \neq \emptyset$. Choose x_n in this set. Then $d(x_n, x) < 1/n \to 0$, so $x_n \to x$.

($\Leftarrow$) If $x_n \to x$ with $x_n \in A \setminus \{x\}$, then for any $r > 0$, $x_n \in B_r(x) \cap (A \setminus \{x\})$ for large n. Hence x is a limit point. □

4.4 Characterization of Closed Sets

Theorem 4.2 (Closed iff Contains All Limit Points)**.** F is closed iff F contains all its limit points.

Proof. ($\Rightarrow$) Let F be closed, so F^c is open. Let x be a limit point of F. If $x \notin F$, then $x \in F^c$. Since F^c is open, $\exists r > 0$ with $B_r(x) \subset F^c$, so $B_r(x) \cap F = \emptyset$, contradicting that x is a limit point. Thus $x \in F$.

($\Leftarrow$) Suppose F contains all its limit points. Take $x \in F^c$. Since x is not a limit point of F, $\exists r > 0$ with $B_r(x) \cap F = \emptyset$, i.e., $B_r(x) \subset F^c$. So F^c is open. □

Theorem 4.3 (Sequential Characterization of Closed Sets)**.** F is closed iff for every sequence $(x_n) \subset F$ with $x_n \to x$, we have $x \in F$.

Proof. ($\Rightarrow$) If $x \notin F$, then $x \in F^c$ (open), so $\exists r > 0$ with $B_r(x) \subset F^c$. But $x_n \to x$ implies $x_n \in B_r(x)$ for large n, contradicting $x_n \in F$. Hence $x \in F$.

($\Leftarrow$) If F does not contain all its limit points, then there's a limit point $x \notin F$. By the sequential characterization of limit points, there's a sequence in F converging to x, contradicting the hypothesis. □

4.5 Properties of Closed Sets

Theorem 4.4 (Arbitrary Intersections)**.** The intersection of any collection of closed sets is closed.

Proof. Let $\{F_\alpha\}$ be closed sets. Then $X \setminus F_\alpha$ is open. By De Morgan:

$$X \setminus \left(\bigcap_\alpha F_\alpha \right) = \bigcup_\alpha (X \setminus F_\alpha),$$

which is open. Hence $\bigcap_\alpha F_\alpha$ is closed. □

Theorem 4.5 (Finite Unions)**.** The union of finitely many closed sets is closed.

Proof. Let $F_1, \ldots, F_n$ be closed. Then $X \setminus F_k$ are open. By De Morgan:

$$X \setminus \left(\bigcup_{k=1}^{n} F_k \right) = \bigcap_{k=1}^{n} (X \setminus F_k),$$

which is a finite intersection of open sets, hence open. So $\bigcup_{k=1}^{n} F_k$ is closed. □

Remark 4.2 (Infinite Unions Fail)**.** $\bigcup_{n=1}^{\infty} [1/n, 1] = (0, 1]$ is not closed.

Theorem 4.6 (Closed Balls are Closed)**.** Every closed ball $\overline{B}_r(x)$ is a closed set.

Proof. We show $X \setminus \overline{B}_r(x)$ is open. Let $y \notin \overline{B}_r(x)$, so $d(x, y) > r$. Let $s = d(x, y) - r > 0$. For $z \in B_s(y)$:

$$d(x, z) \geq d(x, y) - d(y, z) > d(x, y) - s = r.$$

So $z \notin \overline{B}_r(x)$. Thus $B_s(y) \subset X \setminus \overline{B}_r(x)$, proving it is open. □

5 Interior, Closure, and Boundary

5.1 Closure of a Set

Definition 5.1 (Closure)**.** $\overline{A}$ = intersection of all closed sets containing A.

Theorem 5.1 (Closure Equals A plus Limit Points)**.** $\overline{A} = A \cup A'$, where A' is the set of limit points of A.

Proof. Let $B = A \cup A'$. We show B is closed and is the smallest closed set containing A.

Step 1: B is closed. Let x be a limit point of B. We show $x \in B$. Let $r > 0$. $B_r(x) \cap (B \setminus \{x\}) \neq \emptyset$, so there exists $y \in B_r(x) \cap B$, $y \neq x$. Two cases:

- $y \in A$: Then $B_r(x) \cap (A \setminus \{x\}) \neq \emptyset$, so $x \in A' \subset B$.
- $y \in A'$: Since y is a limit point of A and $B_r(x)$ is an open neighbourhood of y, there is $z \in A \cap (B_r(x) \setminus \{y\})$ with $z \neq y$. If $z \neq x$, then x is a limit point of A. If $z = x$, then $x \in A$. Either way, $x \in B$.

So B contains all its limit points, hence B is closed.

Step 2: B is the smallest closed set containing A. Let F be closed with $A \subset F$. If $x \in A'$, then x is a limit point of F (since $A \subset F$). As F is closed, $x \in F$. So $A' \subset F$, hence $B = A \cup A' \subset F$. Combined with Step 1, $B = \overline{A}$. □

Theorem 5.2 (Properties of Closure)**.** (i) $\overline{A}$ is closed.

(ii) $A \subset \overline{A}$.

(iii) $\overline{A}$ is the smallest closed set containing A.

(iv) A is closed iff $A = \overline{A}$.

(v) $\overline{\overline{A}} = \overline{A}$.

(vi) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

(vii) $\overline{A \cap B} \subset \overline{A} \cap \overline{B}$ (may be strict).

Proof. **(i)–(iv)** follow from the previous theorem and definition.

(v) By (i), $\overline{A}$ is closed, so by (iv), $\overline{\overline{A}} = \overline{A}$.

(vi) $A \subset A \cup B$ implies $\overline{A} \subset \overline{A \cup B}$, similarly $\overline{B} \subset \overline{A \cup B}$, so $\overline{A} \cup \overline{B} \subset \overline{A \cup B}$. Conversely, $\overline{A} \cup \overline{B}$ is closed (finite union of closed sets) and contains $A \cup B$, so $\overline{A \cup B} \subset \overline{A} \cup \overline{B}$. Equality follows.

(vii) Since $A \cap B \subset A$, $\overline{A \cap B} \subset \overline{A}$, similarly $\subset \overline{B}$, so $\overline{A \cap B} \subset \overline{A} \cap \overline{B}$. Strictness: $A = (0,1)$, $B = (1,2)$: $\overline{A \cap B} = \emptyset$ but $\overline{A} \cap \overline{B} = \{1\}$. □

Example 5.1 (Closures in $\mathbb{R}$)**.** $\overline{(0,1)} = [0,1]$; $\overline{\mathbb{Q}} = \mathbb{R}$; $\overline{\mathbb{Z}} = \mathbb{Z}$; $\overline{\{1/n\}} = \{1/n\} \cup \{0\}$.

5.2 Boundary of a Set

Definition 5.2 (Boundary Point)**.** x is a **boundary point** of A if every neighbourhood of x meets both A and A^c. The **boundary** ∂A is the set of boundary points.

Theorem 5.3 (Boundary Characterization)**.**

$$\partial A = \overline{A} \setminus \operatorname{int}(A) = \overline{A} \cap \overline{X \setminus A}.$$

Proof. **First identity:** Let $x \in \partial A$. Since every neighbourhood of x meets A, $x \in \overline{A}$. Since every neighbourhood of x meets A^c, $x \notin \operatorname{int}(A)$. Hence $x \in \overline{A} \setminus \operatorname{int}(A)$.

Conversely, if $x \in \overline{A} \setminus \operatorname{int}(A)$: $x \in \overline{A}$ means every neighbourhood of x meets A. $x \notin \operatorname{int}(A)$ means no ball around x is contained in A, so every ball around x meets A^c. Hence $x \in \partial A$.

Second identity: By symmetry, $\partial A = \partial(X \setminus A)$, and applying the first identity to $X \setminus A$: $\partial(X \setminus A) = \overline{X \setminus A} \setminus \operatorname{ext}(A)$. Using the partition $X = \operatorname{int}(A) \cup \partial A \cup \operatorname{ext}(A)$, we get $\partial A = \overline{A} \cap \overline{X \setminus A}$. □

Example 5.2 (Boundaries in $\mathbb{R}$)**.** $\partial(0,1) = \{0,1\}$; $\partial\mathbb{Q} = \mathbb{R}$; $\partial\mathbb{Z} = \mathbb{Z}$.

5.3 Exterior and the Partition of X

Definition 5.3. $\operatorname{ext}(A) = \operatorname{int}(X \setminus A)$.

Theorem 5.4 (Partition of the Space)**.**

$$X = \operatorname{int}(A) \,\dot\cup\, \partial A \,\dot\cup\, \operatorname{ext}(A).$$

Proof. Every $x \in X$ falls into exactly one of the following:

- $x \in \operatorname{int}(A)$: has a ball entirely inside A.
- $x \in \operatorname{ext}(A)$: has a ball entirely inside A^c.
- Otherwise: every ball around x meets both A and A^c, so $x \in \partial A$.

□

6 Bounded Sets and Diameter

6.1 Bounded Sets and Diameter

Definition 6.1 (Bounded Set)**.** A is **bounded** if $A \subset B_r(x)$ for some $x \in X$, $r > 0$.

Definition 6.2 (Diameter)**.** $\operatorname{diam}(A) = \sup\{d(x, y) : x, y \in A\}$.

Theorem 6.1 (Bounded iff Finite Diameter)**.** A is bounded iff $\operatorname{diam}(A) < \infty$.

Proof. ($\Rightarrow$) If $A \subset B_r(x)$, then for $a, b \in A$: $d(a, b) \leq d(a, x) + d(x, b) < 2r$. So $\operatorname{diam}(A) \leq 2r < \infty$.

($\Leftarrow$) If $\operatorname{diam}(A) = D < \infty$, fix $x_0 \in A$. For any $a \in A$, $d(x_0, a) \leq D$, so $A \subset B_{D+1}(x_0)$. □

Example 6.1 (Diameters in $\mathbb{R}$)**.** $\operatorname{diam}((0, 1)) = 1$; $\operatorname{diam}(\mathbb{Q} \cap (0, 1)) = 1$; $\operatorname{diam}(\mathbb{N}) = \infty$.

Example 6.2 (Discrete Metric)**.** $\operatorname{diam}(A) = 0$ if $|A| \leq 1$, and 1 if $|A| \geq 2$.

Example 6.3 (Diameter in ℓ^∞)**.** The set $A = \{e_n : n \in \mathbb{N}\}$ where e_n is the n-th standard basis vector has $d_\infty(e_m, e_n) = 1$ for $m \neq n$. So $\operatorname{diam}(A) = 1$.

6.2 Distance Between Sets

Definition 6.3. $d(x, A) = \inf\{d(x, a) : a \in A\}$; $d(A, B) = \inf\{d(a, b) : a \in A, b \in B\}$.

Remark 6.1. d is not a metric on subsets: $d(A, B) = 0$ does not imply $A = B$.

Theorem 6.2 (Distance to a Closed Set)**.** If A is closed and $x \notin A$, then $d(x, A) > 0$.

Proof. If $d(x, A) = 0$, then for each n there's $a_n \in A$ with $d(x, a_n) < 1/n$. Then $a_n \to x$, so x is a limit point of A. Since A is closed, $x \in A$, contradiction. Hence $d(x, A) > 0$. □

Theorem 6.3 (Distance Function is Continuous)**.** For fixed non-empty A, the map $x \mapsto d(x, A)$ is continuous (indeed, 1-Lipschitz).

Proof. For $x, y \in X$ and any $a \in A$:

$$d(x, a) \leq d(x, y) + d(y, a).$$

Taking the infimum over $a \in A$: $d(x, A) \leq d(x, y) + d(y, A)$. So $d(x, A) - d(y, A) \leq d(x, y)$. By symmetry, $|d(x, A) - d(y, A)| \leq d(x, y)$. □

7 Subspace of a Metric Space

Definition 7.1 (Subspace)**.** For $Y \subset X$, $(Y, d|_{Y \times Y})$ is a **subspace** of X.

Definition 7.2 (Relatively Open)**.** $E \subset Y$ is **relatively open** in Y if it is open in the subspace metric.

Theorem 7.1 (Characterization of Relatively Open Sets)**.** $E \subset Y$ is relatively open in Y iff $E = U \cap Y$ for some open $U \subset X$.

Proof. ($\Rightarrow$) If E is open in Y, for each $y \in E$ there's $r_y > 0$ with $B^Y_{r_y}(y) \subset E$. Let $U = \bigcup_{y\in E} B^X_{r_y}(y)$. Then U is open in X and $U \cap Y = \bigcup_{y\in E} B^Y_{r_y}(y) \subset E$. Also $E \subset U \cap Y$. So $E = U \cap Y$.

($\Leftarrow$) If $E = U \cap Y$ with U open in X, for $y \in E$, $\exists r > 0$ with $B^X_r(y) \subset U$. Then $B^Y_r(y) = B^X_r(y) \cap Y \subset U \cap Y = E$. Hence E is open in Y. □

Theorem 7.2 (Characterization of Relatively Closed Sets)**.** $E \subset Y$ is relatively closed in Y iff $E = F \cap Y$ for some closed $F \subset X$.

Proof. E is relatively closed iff $Y \setminus E$ is relatively open iff $Y \setminus E = U \cap Y$ for open $U \subset X$. Let $F = X \setminus U$, which is closed. Then $E = Y \setminus (U \cap Y) = Y \cap (X \setminus U) = Y \cap F$.

Conversely, if $E = F \cap Y$ with F closed, then $Y \setminus E = Y \cap (X \setminus F)$, and $X \setminus F$ is open. So $Y \setminus E$ is relatively open, hence E is relatively closed. □

Example 7.1. In $X = \mathbb{R}$, $Y = [0, 1]$:

- $[0, 1/2) = Y \cap (-1, 1/2)$, relatively open, but not open in $\mathbb{R}$.
- $[0, 1/2] = Y \cap (-1, 1/2]$, relatively closed.
- $(0, 1) = Y \cap (0, 1)$, relatively clopen in Y.

8 Comparison of Metrics

8.1 Motivation

On the same set X, we can often put many different metrics. The natural question is: *When do two metrics give rise to the same "geometry"?* All properties of a metric space are determined by the family of open sets, so metrics should be considered equivalent when they induce the same topology. However, there are stronger notions of equivalence that preserve more structure.

8.2 Three Notions of Equivalence

Definition 8.1 (Topologically Equivalent Metrics)**.** Two metrics d_1, d_2 on X are **topologically equivalent** if they induce the same open sets, i.e., the identity map $\mathrm{id} : (X, d_1) \to (X, d_2)$ is a homeomorphism.

Definition 8.2 (Uniformly Equivalent Metrics)**.** d_1, d_2 are **uniformly equivalent** if the identity map is uniformly continuous in both directions:

- $\forall \varepsilon > 0, \exists \delta > 0 : d_1(x, y) < \delta \Rightarrow d_2(x, y) < \varepsilon$.
- $\forall \varepsilon > 0, \exists \delta > 0 : d_2(x, y) < \delta \Rightarrow d_1(x, y) < \varepsilon$.

Definition 8.3 (Lipschitz (Strongly) Equivalent Metrics)**.** d_1, d_2 are **Lipschitz equivalent** if there exist constants $c_1, c_2 > 0$ such that for all $x, y \in X$:

$$c_1\, d_1(x, y) \le d_2(x, y) \le c_2\, d_1(x, y).$$

8.3 Hierarchy of Equivalences

Theorem 8.1 (Chain of Implications)**.**

$$\text{Lipschitz equivalent} \implies \text{uniformly equivalent} \implies \text{topologically equivalent.}$$

Neither converse holds in general.

Proof. **Lipschitz ⇒ Uniform:** Suppose $c_1 d_1 \leq d_2 \leq c_2 d_1$. Given $\varepsilon > 0$, choose $\delta = \varepsilon / c_2$. If $d_1(x, y) < \delta$, then $d_2(x, y) \leq c_2 d_1(x, y) < \varepsilon$. So id is uniformly continuous $(X, d_1) \to (X, d_2)$. Symmetrically for the other direction.

Uniform ⇒ Topological: Uniform continuity is stronger than continuity, so the identity map is continuous in both directions, hence a homeomorphism. Thus d_1 and d_2 induce the same topology.

Converse fails: See Example 8.3 and 8.4 below. □

8.4 Invariants of Topological Equivalence

Theorem 8.2 (Topological Invariants)**.** If d_1, d_2 are topologically equivalent, then (X, d_1) and (X, d_2) have the same:

(i) Open sets, closed sets, closure, interior, and boundary of every set.

(ii) Convergent sequences (with the same limits).

(iii) Continuous functions $f : X \to Y$ and $f : Y \to X$.

(iv) Compact subsets.

(v) Connected subsets.

(vi) Dense subsets and separability.

(vii) Homeomorphisms.

Proof. All these are defined in terms of open sets (or their equivalents), which coincide for both metrics. □

Remark 8.1 (What is NOT Preserved Topologically)**.**

- **Boundedness**: On $\mathbb{R}$, $d(x, y) = |x - y|$ and $d'(x, y) = \min(1, |x - y|)$ are topologically equivalent, but $\mathbb{R}$ is d-unbounded and d'-bounded.
- **Completeness**: Same example—$(\mathbb{R}, d)$ is complete but $(\mathbb{R}, d')$ is not.

8.5 Invariants of Uniform Equivalence

Theorem 8.3 (Uniform Invariants)**.** If d_1, d_2 are uniformly equivalent, then in addition to all topological invariants, they have the same:

(i) Cauchy sequences.

(ii) Complete subsets (and (X, d_1) is complete iff (X, d_2) is).

(iii) Uniformly continuous functions $f : X \to Y$.

(iv) Totally bounded subsets.

Proof. **(i)** Let (x_n) be d_1-Cauchy. Given $\varepsilon > 0$, by uniform continuity of id there's $\delta > 0$ with $d_2(x, y) < \varepsilon$ whenever $d_1(x, y) < \delta$. Choose N such that for $m, n \geq N$, $d_1(x_m, x_n) < \delta$. Then $d_2(x_m, x_n) < \varepsilon$.

(ii) Completeness is equivalent to every Cauchy sequence converging. Since both Cauchy and convergent sequences coincide, completeness is preserved.

(iii) If $f : (X, d_1) \to Y$ is uniformly continuous and id: $(X, d_1) \to (X, d_2)$ is uniformly continuous, then f is uniformly continuous from (X, d_2). Given $\varepsilon > 0$, choose $\delta_1 > 0$ with $d_1(x, y) < \delta_1 \Rightarrow d_Y(f(x), f(y)) < \varepsilon$. Choose $\delta_2 > 0$ with $d_2(x, y) < \delta_2 \Rightarrow d_1(x, y) < \delta_1$.

(iv) Similar argument with the definition of total boundedness. □

8.6 Key Examples and Counterexamples

Example 8.1 (d_1, d_2, d_∞ on $\mathbb{R}^n$ are Lipschitz Equivalent)**.** For $x, y \in \mathbb{R}^n$:

$$\|x - y\|_\infty \leq \|x - y\|_2 \leq \|x - y\|_1 \leq n\|x - y\|_\infty.$$

Thus $d_\infty \leq d_2 \leq d_1 \leq n\, d_\infty$, so all three metrics are Lipschitz equivalent. This is why we can use whichever is most convenient.

Example 8.2 (d and $\min(1, d)$ are Uniformly Equivalent)**.** Let $d'(x, y) = \min(1, d(x, y))$. Then:

- $d'(x, y) \leq d(x, y)$, so id: $(X, d) \to (X, d')$ is uniformly continuous with $\delta = \varepsilon$.
- For the other direction, given $\varepsilon > 0$ (assume $\varepsilon < 1$), choose $\delta = \varepsilon$. If $d'(x, y) < \delta \leq 1$, then $d'(x, y) = d(x, y)$, so $d(x, y) < \varepsilon$.

Thus d and d' are uniformly equivalent. They are **not** Lipschitz equivalent if d is unbounded, because d' is bounded.

Example 8.3 (Topologically Equivalent but not Uniformly Equivalent)**.** On $X = \mathbb{R}$, let

$$d_1(x, y) = |x - y|, \qquad d_2(x, y) = |\arctan(x) - \arctan(y)|.$$

Then d_2 is a metric (arctan is injective), and d_1, d_2 are topologically equivalent because $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is a homeomorphism.

They are **not** uniformly equivalent: take $x_n = n$, $y_n = n + 1$. Then $d_2(x_n, y_n) \to 0$, but $d_1(x_n, y_n) = 1$. So id: $(X, d_2) \to (X, d_1)$ is not uniformly continuous.

Example 8.4 (Discrete Metric is not Equivalent to the Usual Metric on $\mathbb{R}$)**.** The discrete metric induces the discrete topology (every subset open), while the usual metric induces the standard topology. Since $\{0\}$ is open in the discrete metric but not in the usual one, they are not topologically equivalent.

8.7 Criteria for Equivalence

Theorem 8.4 (Sufficient Condition for Topological Equivalence)**.** If for every $x \in X$ and every $r > 0$, there exist $r_1, r_2 > 0$ such that

$$B_{r_1}^{d_1}(x) \subset B_r^{d_2}(x) \quad \text{and} \quad B_{r_2}^{d_2}(x) \subset B_r^{d_1}(x),$$

then d_1 and d_2 are topologically equivalent.

Proof. Under the hypothesis, every d_1-open set is d_2-open (use the first inclusion) and vice versa. Hence the topologies coincide. □

Theorem 8.5 (Restriction to Subspaces)**.** If d_1, d_2 are topologically (resp. uniformly, Lipschitz) equivalent on X, then their restrictions to any $Y \subset X$ are also topologically (resp. uniformly, Lipschitz) equivalent on Y.

Proof. Immediate from the definitions. □

8.8 Summary Table of Equivalences

Notion	**Definition**	**What is Preserved**
Topologically equivalent	Same open sets (homeomorphic identity)	Open/closed sets, closure, interior, boundary, convergence, continuity, compactness, connectedness, separability
Uniformly equivalent	Identity uniformly continuous both ways	All of the above, plus: Cauchy sequences, completeness, uniform continuity, total boundedness
Lipschitz equivalent	$c_1 d_1 \leq d_2 \leq c_2 d_1$ for constants $c_1, c_2 > 0$	All of the above, plus: boundedness, Hölder continuity

9 Tutorial / Problem-Solving Session

9.1 Key Theorems to Master

1. Variant triangle inequality: $|d(x,z) - d(y,z)| \leq d(x,y)$.
2. Open ball is open; closed ball is closed.
3. Arbitrary unions, finite intersections of open sets.
4. Arbitrary intersections, finite unions of closed sets.
5. Closed iff contains all limit points.
6. Closure = $A\cup$ limit points.
7. Boundary = $\overline{A} \setminus \operatorname{int}(A)$.
8. Bounded iff finite diameter.
9. Subspace characterization: relatively open iff $U \cap Y$.
10. $\ell^p \subsetneq \ell^q$ for $p < q$.
11. Hierarchy: Lipschitz $\Rightarrow$ uniform $\Rightarrow$ topological.

9.2 Practice Problems

1. Prove that the union of any collection of open sets is open.
2. Prove that a finite intersection of open sets is open. Show that infinite intersections need not be open.
3. Prove that every closed ball is a closed set.
4. Show that in a metric space, $\overline{A} = A \cup A'$.
5. In $\mathbb{R}$, find the interior, closure, and boundary of $(0,1) \cup \{2\}$ and $\mathbb{Q}$.
6. Describe all open sets, closed sets, and limit points in a discrete metric space.
7. Let $Y = [0,1]$ as a subspace of $\mathbb{R}$. Show $[0, 1/2)$ is relatively open and $[0, 1/2]$ is relatively closed.
8. Prove that $\ell^1 \subsetneq \ell^2 \subsetneq \ell^\infty$.
9. Prove that d_1, d_2, d_∞ on $\mathbb{R}^n$ are Lipschitz equivalent.
10. Show that d and $d/(1+d)$ on any metric space are uniformly equivalent.
11. Give an example of two metrics on $\mathbb{R}$ that are topologically equivalent but not uniformly equivalent.
12. Prove that completeness is preserved under uniform equivalence but not under topological equivalence.
13. Draw the unit balls of d_1, d_2, d_∞ on $\mathbb{R}^2$ and identify the point $(0.8, 0.6)$ on each.
14. Let ℓ^∞ denote the space of bounded sequences. Show ℓ^∞ with d_∞ is a metric space.

Summary of Key Concepts

Concept	Definition	Key Property
Metric	$d : X \times X \to [0, \infty)$ satisfying positivity, symmetry, triangle inequality	Abstracts "distance"
Open Ball	$B_r(x) = \{y : d(x, y) < r\}$	Always an open set
Closed Ball	$\overline{B}_r(x) = \{y : d(x, y) \leq r\}$	Always a closed set
Open Set	Every point has a ball contained in the set	Arbitrary unions, finite intersections
Closed Set	Complement is open	Contains all limit points
Limit Point	Every neighbourhood contains a point of the set different from it	Characterizes closed sets
Isolated Point	Point in set that is not a limit point	Has a neighbourhood containing only that point
Interior	$\text{int}(A) = \{x : \exists r > 0, B_r(x) \subset A\}$	Largest open subset of A
Closure	$\overline{A} = A \cup \{\text{limit points}\}$	Smallest closed set containing A
Boundary	$\partial A = \overline{A} \setminus \text{int}(A)$	Points close to both A and A^c
Bounded Set	Contained in some ball	Finite diameter
Diameter	$\text{diam}(A) = \sup\{d(x, y) : x, y \in A\}$	Measures "size" of set
Subspace	$Y \subset X$ with restricted metric	Relative open/closed sets
ℓ^p containment	$1 \leq p < q \leq \infty \Rightarrow \ell^p \subsetneq \ell^q$	Higher p is more permissive
Topologically equivalent	Same open sets	Same topology, same continuity
Uniformly equivalent	Identity uniformly continuous both ways	Same Cauchy sequences, same completeness
Lipschitz equivalent	$c_1 d_1 \leq d_2 \leq c_2 d_1$	Same bounded sets, same Lipschitz structure

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